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Bachelor Thesis

Data Visualization for Augmented Reality Head-Up Displays in Personal Protective Equipment

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1 Introduction

Despite rapid advances and numerous potential applications for augmented reality, it has not yet been widely adopted in industrial settings. One such area of application is in activities where chemical protective clothing is needed. This inherently dangerous work involves significant physical exertion. These activities require constant awareness of the environment and access to information, which is provided by monitoring devices and radio communication in current practice.

The protective equipment restricts the senses and limits mobility, which makes every movement a challenge. A head-up display is able to integrate relevant information into the field of view, thereby enabling hands-free and immediate access, potentially replacing current methods.

However, workers are already pushed to their limits by the equipment and task demands. The HUD must not introduce additional burden—neither in terms of weight nor cognitive load—and it must be possible to utilize the system without interrupting current tasks.

First, the requirements of the intended use case, the limitations imposed by AR technology, and biological constraints are identified. Next, the mechanisms of attention are introduced, including how they can be utilized to design a head-up display capable of providing relevant information without causing distractions. This is followed by a compilation of design options that enable rapid access to and comprehension of information within the aforementioned constraints. Finally, design guidelines are proposed that compile the results of studies on AR and attention management and are illustrated using a potential HUD design.

2 Chemical Protective Clothing and Its Applications

Scenarios where workers are exposed to hazardous materials require chemical protective clothing (CPC). Protection levels range from limited spray protection (Type 6) to fully encapsulating, gas-tight suits (Type 1) with self-contained breathing apparatus (SCBA)[1] ["DGUV Information 212-019"].

Figure 1: Type 1 chemical protection suit 4005/NEO manufactured by isotemp[2]

Confined spaces are defined as areas in which, through their design or site conditions, "inadequate air replacement rates, or the substances [...] found within them or introduced into them, particular hazards exist or may arise that substantially exceed the normal prevailing hazard potential of workplaces" ["DGUV Regel 113-004"]. Tasks that require entering confined spaces are typically cleaning, inspection and/or maintenance of, for example, tanks in chemical plants, rail cars and highway tank trailers and even sewage containers. Traces of the contents may remain, even if their containers are prepared correctly. These residues might—

[1] A self-contained breathing apparatus consists of a respirator connected to a gas cylinder supplying breathable air. The cylinder is fastened to a harness worn on the back.

[2] Vollschutzanzug 4005/NEO · Chemie · isotemp

despite being initially inactive—become dangerous through work procedures (e.g. welding), agitation or chemical reactions ["DGUV Regel 113-004"], hence requiring the highest levels of protection. Other use cases are emergencies, such as confined space rescue, removing chemical spills or other accidents involving unknown, possibly hazardous substances ["OSHA Technical Manual Section VIII"].

These tasks are highly demanding—both physically and mentally—particularly because of the equipment. Since Type 1 suits are not gas-permeable, they compromise the body's ability to regulate temperature, leading to exhaustion and heat fatigue even faster [Holmér, 2006]. The exertion is exacerbated by heavier equipment and heat stress, ultimately reducing work capacity and performance [Smolander et al., 1984]. The maximum working time in a full-body suit with SCBA is therefore limited and might decrease even more depending on oxygen and exhaustion levels.

Further, the material and shape of the suit limit mobility and the field of view to such an extent that even trained personnel may take twice as long to complete tasks when wearing CPC [Murray et al., 2011].

3 Augmented Reality

Augmented reality (AR) is a subcategory of extended reality (XR), alongside mixed reality (MR) and virtual reality (VR). In AR, virtual objects are "superimposed upon or composited with the real world" [Azuma, 1997], in opposition to virtual reality which—by replacing the real environment entirely—creates the deepest sense of immersion. Mixed reality combines features of both concepts, the main distinction being the ability to interact with virtual objects in the real world [Hari Chandana et al., 2023].

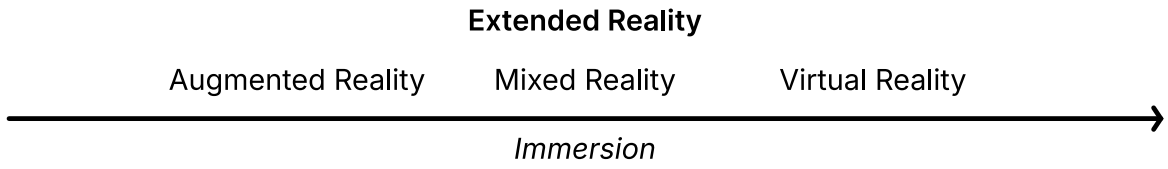

Figure 2: The extended reality spectrum, encompassing augmented, mixed and virtual reality, sorted by degree of immersion, from low to high.

Azuma (1997) defines three requirements for applications to be classified as AR. They combine real and virtual content, they are interactive in real-time, and the virtual objects are registered in 3D space. With the technological advancements since the 2000s these requirements are understood more as guidelines rather than rules.

Head-up displays (HUDs) overlay information onto reality in the user's line of sight, allowing them to access necessary information without having to look away from their primary task. Originally used in military aircraft, today HUDs are installed into consumer vehicles as well. In addition to these integrated HUDs, content can be presented on a head-mounted display. In this case, the display is attached to the head, typically as a helmet or glasses, enabling mobile usage. In contrast to the above definition by Azuma (1997), the content displayed on a HUD is two-dimensional and head-stabilized, rather than spatially anchored, i.e., world-stabilized (see Figure 3). Strictly speaking, a HUD therefore does not meet the criteria for AR. However, other research considers head-stabilized presentation of content as suitable for augmented reality [Feiner et al., 1993, Lu et al., 2020].

G. Lin et al. (2021) also point out that world-stabilized graphics are rarely used in industrial applications, particularly in full-shift scenarios where the HMD would have to be worn continuously. Furthermore, work tasks often require an overview of the situation rather than spatially distributed information. Additionally, with a 2D

HUD, elements are always displayed in the same location, making it easier to find the needed information.

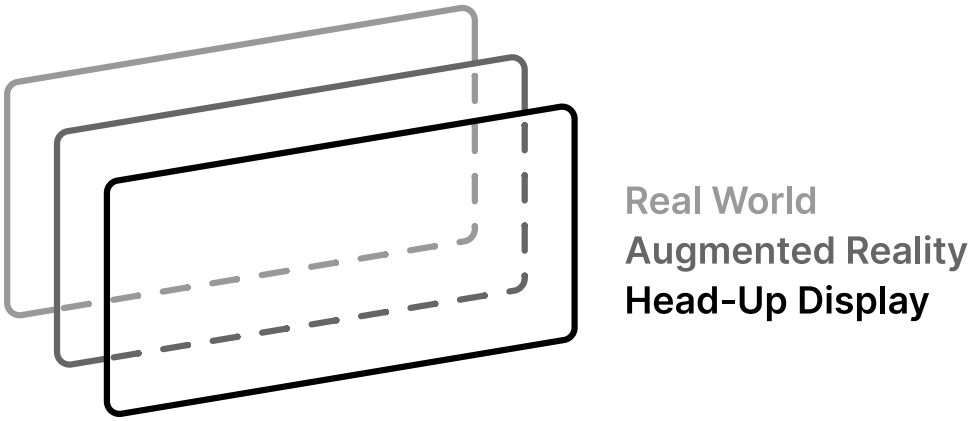

Figure 3: Visualization layers of head-mounted displays: reality, Augmented Reality (3D, world-stabilized) and User Interface (2D, head-stabilized)

3.1 Optical See-Through Displays

There are two approaches to combining virtual content with reality. With video see-through displays, information is superimposed onto a video stream captured by a camera mounted on the device. These headsets fully enclose the head to block external light. In contrast, an optical see-through (OST) system allows the user to view reality directly. Here, the overlay is created by optical combiners blending light from the image source and the environment together.

Since video see-through displays completely block the view of the outside world, it is impossible to see anything in the event of a failure. Given the intended use cases, this represents a major safety issue, especially since the headset cannot be removed while wearing a fully encapsulating suit.

3.1.1 Ambient Light and Contrast Issues

Regardless of the optic technology used, images are created by combining light, which means the display has to compete with ambient light for visibility. This results in a significant loss of contrast, especially outdoors. Kerr et al. (2011), for example, report that participants repeatedly used their hands to shield the display from external light to be able to read its contents.

Many manufacturers of optical see-through displays equip them with shaded lenses to improve the contrast somewhat [Itoh et al., 2021]. Unfortunately, current OST technology is incapable of producing the contrast levels required for outdoor use [Erickson et al., 2020]. Technical solutions currently in development include OSTs that can partially occlude areas of the display to block light. However, these are still very experimental [Kiyokawa et al., 2000, Davari et al., 2020].

As a hardware-independent solution, Tanaka et al., 2008 propose a method for automatically placing content where readability is better, i.e. darker segments of the background. Such concepts might not be suitable for safety-critical scenarios, as users rely on spatially anchored elements to quickly acquire information.

3.1.2 Color Blending

In the additive color model, the combination of all wavelengths of light produce white, while black is created by the absence of light. When physical screens display black, the corresponding pixels do not emit light, so the actual color of the display material is visible. Since OHMDs do not have a physical screen, "black" is transparent [Gabbard et al., 2013]. This also means that darker colors are not available for design on OHMDs, as they are mostly translucent. Additionally, the level of transparency affects the extent to which the background distorts the color [Itoh et al., 2021].

Gabbard et al. (2013) additionally found that, depending on the background, colors may shift in hue or washout due to their luminance value and/or chromaticity. Both of these effects result in colors potentially being misinterpretated or not being visible at all.

4 The Human Visual System

4.1 The Field of View and Peripheral Vision

The human field of view spans approximately 200° to 220° horizontally and around 125° vertically, with the vertical field of view extending slightly further downward [Loschky et al., 2017]. The central vision covers about 5° around the point of fixation, where acuity is highest [Rosenholtz, 2016, Loschky et al., 2017]. The remaining field is the periphery, with boundaries at 30° and 60°, dividing it into near, mid and far peripheral vision.

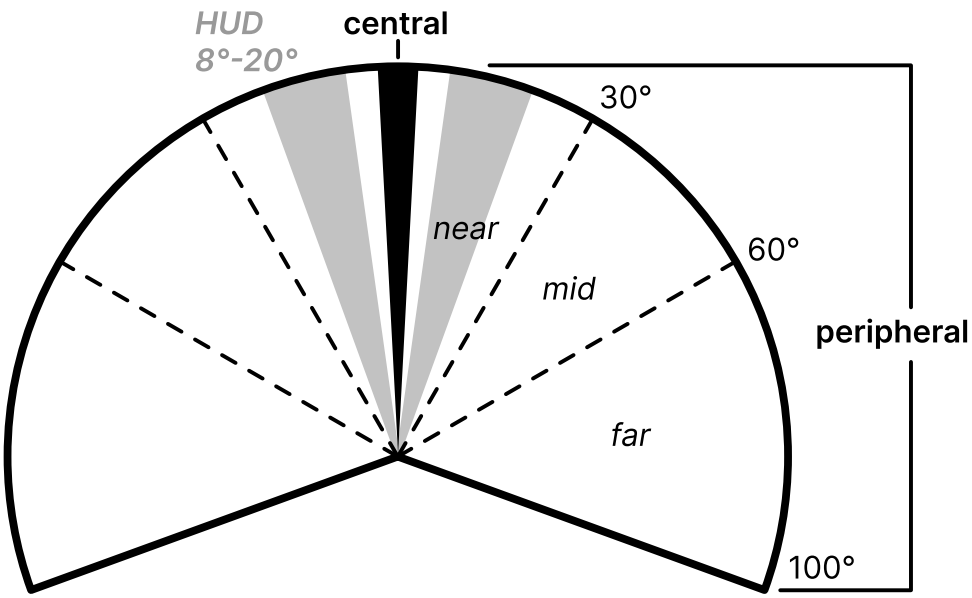

Figure 4: Simplified illustration of the horizontal field of view

Despite visual acuity and other functions deteriorating in the periphery, the area in front of us continues to appear in high resolution, with high contrast and color saturation [Loschky et al., 2017]. According to Rosenholtz (2016), the decline in performance of the peripheral vision is smaller than commonly assumed. A scene does not require many details to still allow object recognition and orientation, which is why image thumbnails remain recognizable despite their small size.

In addition, peripheral vision is particularly well suited for detecting motion, provides better vision in low-light conditions [Johnson, 2014] and contributes to functions like orientation, navigation, balance, and hand-eye coordination. A restriction of the peripheral visual field severely impairs the performance of these functions, the effect increasing with the amount of vision obstructed [Alfano and Michel,

1990].

Color sensitivity decreases toward the periphery, which is more pronounced for red-green colors than for blue-yellow colors and for brightness perception. Hansen et al. (2009) found that color perception is possible even at an eccentricity of 50°, provided the stimulus is sufficiently large. Most modern OST-HMDs feature a 40° field of view (see Section 3.1), meaning the entire available color spectrum can be utilized.

Rosenholtz (2016) suggests that crowding is the biggest issue in parsing information located in the periphery. When visual elements are placed too close together, they blend into a single entity, making it harder to distinguish them from one another. This effect is more pronounced if the target and its neighboring elements are similar (e.g., same color and shape) [Pelli and Tillman, 2008]. The farther elements are from the center of vision, the greater the distance between them must be.

4.2 Visual Attention

The retina transmits too much information to be processed all at once. This bottle neck is resolved by visual attention, which focuses resources on just a fragment of the current visual stimuli [Anderson et al., 2005, Rosenholtz, 2016].

Two processes determine where this attention is focused next. Bottom-up attention occurs involuntarily, triggered by external stimuli. In the top-down process, the gaze is directed toward objects related to current needs or goals. Interestingly, top-down attention reduces the impact of distracting stimuli while consciously focusing on a visual target [Connor et al., 2004].

An object is referred to as visually salient, if it attracts attention by being distinct from its surroundings in some way, without specifying which characteristics cause it to do so. When multiple stimuli compete for attention, even salient targets might be ignored if they are irrelevant to the current goal. This is referred to as inattentional blindness [Jensen et al., 2011].

5 Attention Management

After discussing the mechanisms of perception and attention, the question arises as to how they can be utilized and which aspects need to be taken into consideration in order to do so.

5.1 Situation Awareness

Making good judgements becomes increasingly difficult when performing demanding procedures while trying to maintain awareness of the surrounding area. The risks associated with these activities prohibit human error, further increasing stress levels. Endsley (1995) highlights the role of situation awareness as a foundation for good decision-making in dynamic situations. She outlines three levels of SA, namely perception, comprehension and projection, of which perception is the most significant for UI design. This stage requires collecting the information relevant to the situation. In level 2 this information will be analyzed to form "a holistic picture of the environment, comprehending the significance of objects and events." Finally, likely outcomes can be determined by understanding the dynamics of the situation.

It is evident that situation awareness is therefore related to the processes of attention described in the previous section. Subsequently, this illustrates the necessity of a thorough understanding of the pieces required for achieving good situation awareness to design a well-functioning HUD. The amount of information that can be rapidly and accurately obtained is directly influenced by how it is presented. In addition to prioritizing and organizing that information around task objectives, Endsley (1995) mentions the advantage of systems providing processed information rather than the data it is based on (e.g. displaying remaining time based on current oxygen amount and usage).

She emphasizes that special attention should be paid to reserve highly salient features to critical events that truly necessitate changing the course of action. Further, it is vital to maintain a balance between showing all relevant information, while simultaneously avoiding visual overload. While looking for inaccessible information, other important content may be overlooked. Conversely, "filtering of extraneous information (not related to SA needs) and reduction of data (by processing and integrating low-level data to arrive at SA requirements)" is necessary to prevent clutter. This reiterates the importance of accurately evaluating and prioritizing

task-relevant information.

As Endsley (1995) points out, considering these recommendations supports assessment of the interaction of all components as a whole, since emphasizing some elements inevitably comes at the expense of others. Evaluating the design concept by the amount of situation awareness it provides serves as the last step of ensuring the effectiveness of the system.

5.2 Peripheral Displays and Glanceability

The requirements of situation awareness make clear that information has to be quickly accessible, without straining cognitive capacity or being overly distracting. Peripheral displays do just that—keeping relevant information accessible while offering the choice of whether or not to continue with the current task [Matthews et al., 2005]. On the other hand, so-called notification displays actively interrupt the user to prompt necessary action.

Building onto this categorization, Matthews et al. (2006) introduces the concept of glanceability—a term used to describe information that is quickly and easily comprehended once it is paid attention to. Glanceability is directly impacted by the displayed amount of items, their salience, etc.—i.e. the attentional limits described in Section 4. In this context, Matthews et al. (2006) emphasizes that visual components need to "balance distinctiveness with too much demand for attention."

Lu et al. (2020) note that to design effective glanceable displays, it is necessary to consider both the visual presentation of information and how it is accessed. They present two novel concepts for hands-free interaction that implement the principles of glanceability and compare their performance to a regular HUD. Participants in the study were asked to follow a virtual mannequin at a specified distance for the primary task, while being asked questions about the information shown on the display.

The eye-glance interface corresponds to a conventional HUD, in which elements are fixed at the edges of the display's field of view. Although continuously displaying the information makes it easily accessible, it can lead to information overload or otherwise be distracting. Additionally, restricting the peripheral field of view can impair perception in that area.

The visual components of the head-glance interface are placed just outside the

display's field of view and are accessed by turning the head. Without this movement, the elements are not visible at all. This approach requires an additional sensor to track the head movement relative to the position of the body. According to Lu et al. (2020), the interface caused frustration among the participants. However, they theorize that this was due to the frequency of the questions asked, which led to numerous head movements within a short amount of time. In addition, participants found it mentally taxing to remember the location of specific information.

In the gaze-summon interface, visual targets are placed near the position where the interface elements would be located. If these points are fixated for a certain amount of time, the information appears at that location. However, in industrial environments this delay may not be justifiable in the intended applications. Additionally, Lu et al. (2020) report that keeping the gaze on target might lead to eye strain if the display is used frequently. There was also an increase of false positives when participants took a turn, caused by reflexively looking into the intended direction, combined with eye-tracking jitter from the rapid head movement.

These methods offer an interactive approach to countering information overload. That said, visual elements must also be designed in such a way as to draw the user's attention to the relevant information.

5.3 Capturing Attention

Peripheral displays are designed to enable passive monitoring of information. In critical situations, however, the display must be able to capture the user's attention, which can be achieved through the intentional use of motion.

In their "Peripheral Displays Toolkit" Matthews et al. (2004) demonstrate how to assign appropriate notification levels to information and what the transitions between these levels should look like. Abrupt transitions, for example, are meant to interrupt the user in order to attract their full attention. When aiming for divided attention, it should be sufficient to update smaller areas of the screen or to use slower animations.

Janaka, Zhao, and Chan (2023) implemented this approach and demonstrated that using fade-in animations reduces the impact of interruption. The effectiveness of the notifications depends on their duration, position, as well as the complexity of the main task.

6 Data Visualization

"One of the greatest benefits of data visualization is the sheer quantity of information that can be rapidly interpreted if it is presented well" [Ware, 2013].

Data visualization transforms data into information. Data itself is just noise, whereas information holds value—value that is created by the context through which the data is interpreted.

6.1 Guiding Visual Search

Information needs to be located before it can be interpreted. To make this happen as quickly as possible, certain design principles can be utilized.

To this end, Shao et al. (2023) present the 3V method, which refers to visual hierarchy, weight, and flow respectively. The visual weight is described by the features of and space occupied by elements. It works through the distinction of objects, achieved by differing presentation. More visual weight means the object attracts more attention. The visual connections between elements describe the visual flow. If implemented correctly, flow guides the gaze to target information.

In summary, even when a user is consciously (i.e., top-down) searching for a specific element in the user interface, a well-structured design can leverage bottom-up attention to guide this visual search [Shao et al., 2023].

6.2 Facilitating Rapid Comprehension

Despite the majority of data visualization being tailored to complex datasets, some concepts can still be utilized to make information glanceable.

Wilke (2019) identifies position, shape, size and color as the fundamental elements for visualization. Position describes the arrangement of data in relation to other points or axes in a diagram. As for color, a distinction should be made between hue and brightness, as these are appropriate for different types of data. Hues are better suited for clearly defined categories, while brightness is best for continuous data, since the eye is more sensitive to differences in brightness and can therefore distinguish smaller variations.

Each of these elements can be used to encode information. Which element is used for which information influences the type of visualization. Importantly, this mapping must be unique, else it becomes ambiguous and obscures the meaning [Wilke, 2019].

6.3 Visualizations for Single Value Data

Aside from design fundamentals, most specialized resources discuss visualization types intended for interrelated data. Interpreting such complex information requires time and cognitive capacity unavailable in scenarios such as those mentioned in Section 2. In these situations, a clear picture of the current status must be available at all times, which only allows for individual data values and therefore limits the number of possible visualization types.

Ferdio, an agency specialized on infographics and data visualization, has created a library of visualization types[3]. In this library, out of a 166 options, only nine are for single data (see Figure 5).

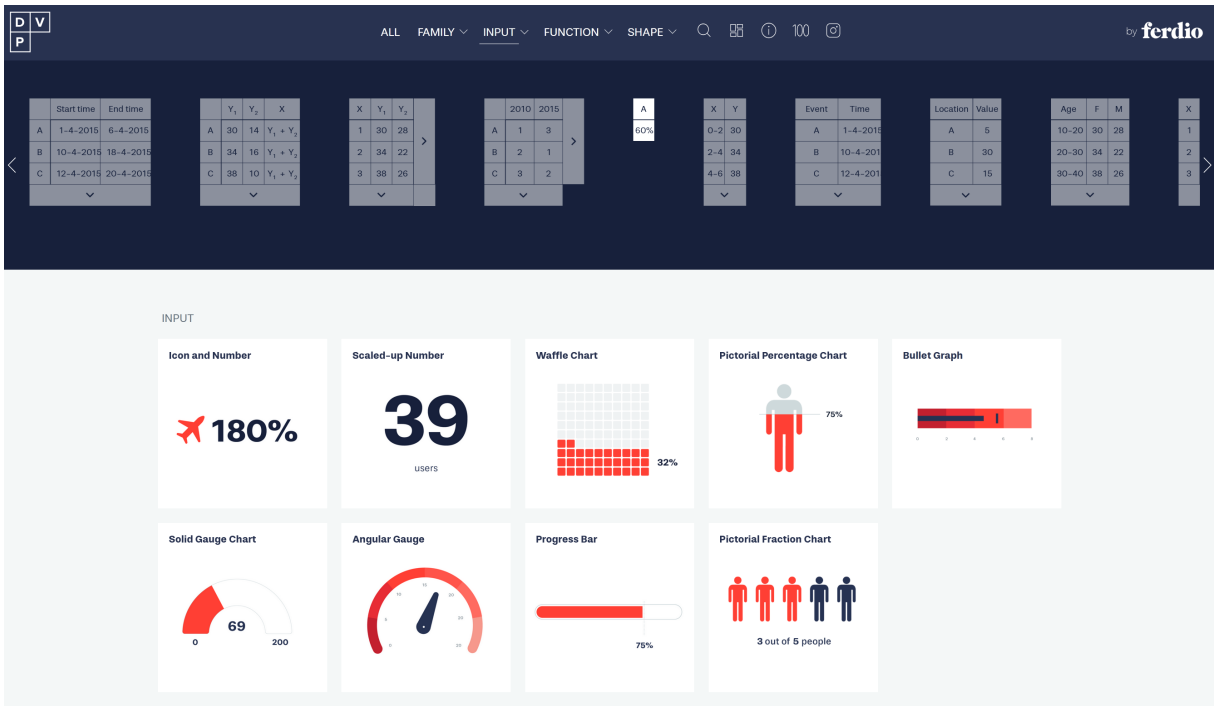

Figure 5: The options for single data (highlighted in the menu bar) in Ferdio's data visualization library.

[3]Data Viz Project | Collection of data visualizations to get inspired and find the right type

7 Design Guidelines

The following guidelines summarize the points discussed in the preceding sections as well as recommendations from other authors. They serve as a basis for the design of head-up displays aiming to support the efficient completion of objectives without compromising safety.

To facilitate situation awareness, the HUD should be capable of providing as much information as possible without causing fatigue or sensory overload. The displayed information should be comprehensible in the shortest possible time. In an emergency, the system must be able to capture the user's attention without hindering decision-making or completely interrupting the primary task.

7.1 Color and Contrast

As mentioned in Section 3, the use of colors in HMDs intended for daylight use is primarily limited by current optical see-through technology. Gabbard et al. (2022) recommend to prefer colors with high source luminance, as these are less prone to color shifts. Furthermore, if a color typically represents a specific meaning and cannot be guaranteed to be perceived correctly, alternative colors that can meaningfully convey that message should be considered. Red is specifically cited as an example, as it is highly susceptible to color shifts due to its low luminance.

The library chroma.js[4] has a function to sort colors by luminance in the RGB color space, based on the Web Content Accessibility Guidelines 2.0.[5] Sorting the base colors red (#FF0000), green (#00FF00), blue (#0000FF) and their intermediaries cyan (#00FFFF), magenta (#FF00FF), and yellow (#FFFF00) puts them into the following order (brightest to darkest): yellow, cyan, green, magenta, red, and blue.

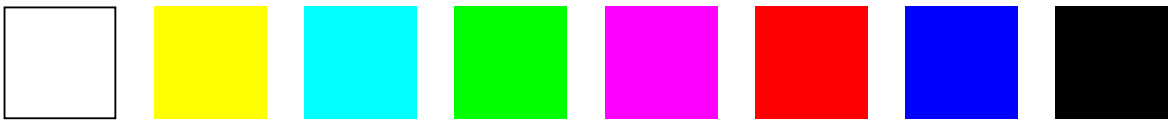

Figure 6: Colors sorted by highest to lowest luminance, in order: yellow, cyan, green, magenta, red, and blue. White and black have been added for reference.

[4] chroma.js api docs!
[5] Web Content Accessibility Guidelines (WCAG) 2.0

Limiting the UI to a single color can ensure that the design functions effectively in every respect (e.g., by using motion to draw attention rather than color). Not encoding information through colors also avoids potential semantic misinterpretations that may be caused by the previously mentioned color issues. In addition, monochrome displays are more affordable and therefore offer economic advantages, since replacement costs should be kept as low as possible in cases where equipment may be damaged or contaminated.

7.2 Typography

As with all head-mounted devices, walking causes them to shake, which affects legibility [Janaka, Zhao, and Chan, 2023]. Especially thin, horizontal lines tend to disappear optically due to the up- and down motion. Matsuura et al. (2019) confirm that heavier font weights improve readability. Naturally, the same requirements apply for stroke width in general.

For example, the typeface "Inter" meets these criteria well. It features a uniform stroke width, a square letter form and good line spacing. Compare this with the typeface "League Gothic", which also has a consistent stroke width, but due to its thickness, there is less white space within each character. Even with increased tracking, the narrowly shaped letters make reading considerably more difficult.

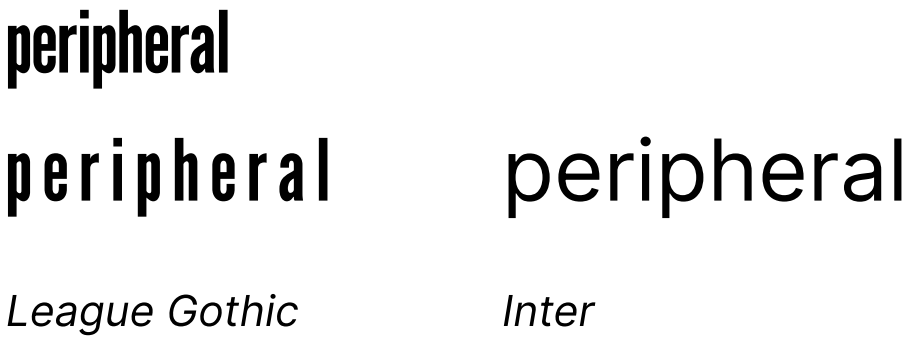

Figure 7: A comparison of the legibility of the typefaces "Inter" and "League Gothic," both set at a size of 32 px and in the regular weight. For "League Gothic," a line with increased character spacing was added, demonstrating that the font remains difficult to read.

To determine optimal font sizes, Renkewitz et al. (2008) tested to what extent the background and font colors affect text recognition. They set a benchmark of two seconds as threshold for fast recognition, which resulted in recommended font sizes from 23 to 42 pt. However, these values may have to be adjusted for use on

an OHMD, as they used a video see-through system and the colors red and blue, both of which have low luminance.

Gattullo et al. (2015) tested different text drawing styles in an indoor industrial setting. When the use of specific colors is mandated (e.g., in accordance with standard signage regulations), they recommend using white text on a billboard in the required color, which generally performed best. However, the billboard style obstructs much of the background. To remedy this, an outline style could be used, with thicker outlines further improving readability.

Figure 8: Billboard and outline text drawing styles

7.3 Iconography

Janaka, Zhao, and Sapkota (2023) tested if, and under which conditions icons perform better than text. They found that icon-augmented notifications enabled significantly higher primary task performance than text only and reduced the level of interruption while maintaining reaction and comprehension. They also found icons are more comprehensible when task complexity increased or demanded more attention. Additionally, the notification were "easier to perceive in suboptimal conditions", including outdoors when walking.

Naturally, this requires the icons to be easily recognizable. To that end, H. Lin et al. (2016) tested different versions of icons against each other. The differences tested were polarity, plane- and shape-based, and the presence or absence of borders. Since line-based icons dissect the space more frequently, they increase visual complexity, which impairs recognition performance. Overall, white shape-based icons on black background performed best.

Figure 9: Line-based and plane-based icons

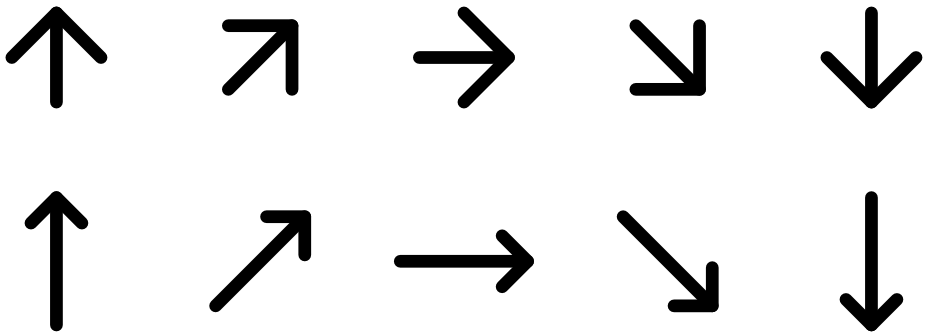

Figure 10: Comparison of short and long arrows.

Lindberg and Näsänen (2003) point out that the size of icons has a greater impact on visual search performance than the spacing between them. However, a spacing of 0.5 to 1 symbol should be maintained. They additionally mention that a grid of 5 $\times$ 5 icons can be processed in one fixation.

7.4 Layout and Spacing

The ability of (binocular) head-mounted displays to display content anywhere within the field of view always presents a trade-off between immediate access to information and an unobstructed central line of sight. In use cases involving a great deal of physical activity, a clear field of view should of course be given greater priority.

Chua et al. (2016) investigated the positioning of content on a monocular OST-HMD worn in front of the right eye. The display was divided into a 3 $\times$ 3 grid, spanning 12.5° vertically and 17° horizontally in both directions from the center. The top and most peripheral positions were found to be more comfortable and unobtrusive, although middle and bottom center positions were perceived more quickly. The center-right was recommended as the best balance between both usability and comfort. They mention that the ideal display position is highly dependent on task requirements (e.g., the center is preferable when driving, while secondary information is better placed in the periphery).

G. Lin et al. (2021) divide HMD applications into four task categories: single (continuously focusing on display), alternating (switching back and forth between display and reality), background (passively monitoring the display for notifications) and dual (displayed information supplements physical primary task, e.g., live captions). Similar to Chua et al. (2016), they mention that optimal positioning depends

on task type. Participants preferred a horizontal angle of 5.4° to 14.6° and found angles beyond that to be uncomfortable.

Dowell et al. (2002) mention 8° around the line of sight should be kept free, since it "can induce cognitive tunneling and impair performance on the task over which the HUD is superimposed." Ultimately, 8 to 20° horizontally seem to be a good compromise for most purposes [G. Lin et al., 2021].

7.5 Notifications

Notification levels should be assigned according to the level of attention or action required. Each level should be coded as clearly as possible by choosing appropriate methods, such as abrupt movements for high urgency. If there are multiple independent high-priority notifications, they could be represented using different types of movement, such as flashing or bouncing. Furthermore, the more elements that attract attention, the less effective they are.

Although the billboard style occludes a large portion of the background around the text—which should generally be avoided—it is precisely this characteristic that makes it suitable for high-priority notifications.

8 Conceptualization

For printing, the color of the designs was inverted to black on white. A UI example in original colors is included as Figure 21.

8.1 Components

A prioritization of elements is necessary in order to establish a visual hierarchy and determine which elements should be omitted if there is not enough space.

	Display Mode	Visualization
Oxygen	Continuous	Status Bar, Value
Hazard Proximity	Pop-Up	Icon
Warning Notification	Pop-Up	Billboard
Working Time	Pop-Up, Continuous	Countdown
Heart Rate	Continuous	Icon, Value, Trend
Ambient Temperature	Continuous	Icon, Value, Trend
Humidity	Continuous	Icon, Value
Checklist	Continuous	Line Highlight
Tool Help	Pop-Up	Icon, Value

Table 1: Parameters chosen for visualization, ordered from most to least important. Continuous means that the component will be displayed permanently. Parameters marked with pop-up will be displayed occasionally when relevant.

8.1.1 Visualizing Oxygen Usage With a Status Bar

Since the oxygen value is the decisive factor whether or not to continue work, this component takes center stage. As the most important parameter, this component will be visible in any version of the HUD. To ensure the current status is recognizable at a glance, the value is visualized with a status bar.

A radial version of the component was created based on mechanical gauges, whose hands indicate the status by default. This component also occupies a rela-

tively square area within the grid, which allows for alternative layout options. However, compared to the linear version, it requires significantly more space to ensure the information is legible. Furthermore, as mentioned in section 6, a linear design enables much faster information processing.

Figure 11: Radial and linear concepts for the status bar

In the second version, the value is displayed to the right below the bar. As the oxygen level decreases, the empty sections of the bar appear only as outlines, which increases the contrast between the value and the bar. The label is placed in front of the bar so that the two blend together, making the label less noticeable. In addition, the status bar has been divided into quarter segments, allowing for a more accurate assessment of the remaining amount without having to look at the value.

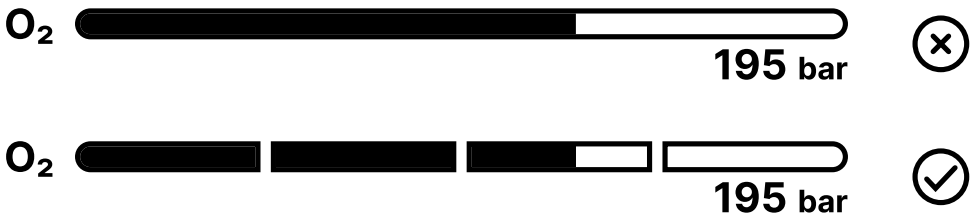

Figure 12: The status bar divided into quarter segments, allowing status assessment without having to look at the value.

Depending on the available screen space and the layout configuration, a vertical version of the component should be available in addition to the horizontal one. Grouping the label and value on the same side saves space. Aligning them at the top and bottom mitigates visual crowding and accentuates the value. The labels are positioned so that the status bar is perceived first, i.e. when the component is placed on the left side, the bar will be on the right to provide visual guidance.

Figure 13: Development of the vertical status bar. The correct version accounts for visual crowding and accentuates the value.

8.1.2 The Base Component

Secondary data—which, while helpful, is not a priority and therefore does not need to be visualized—can be based on a standardized base component. This includes, for example, data such as heart rate, as well as environmental information such as ambient temperature or humidity.

Apart from the actual parameter value, any element of the base component can be hidden. This allows for flexible configuration depending on the type of information. Because they share a common base, the information is visually similar enough to be grouped into blocks.

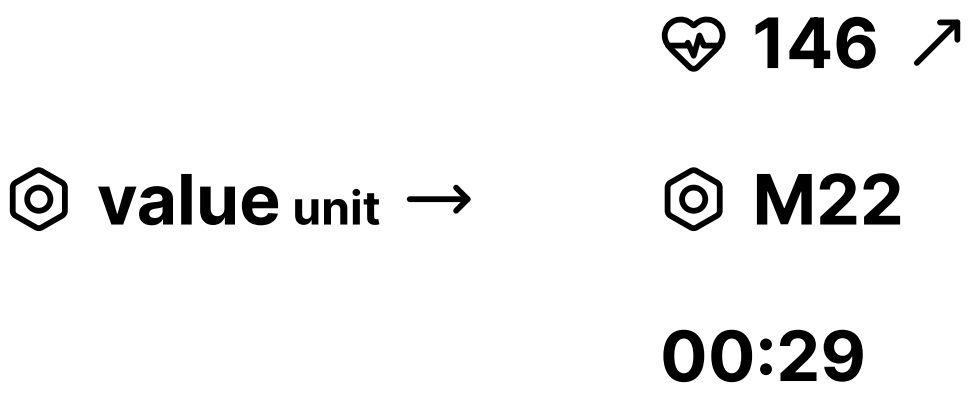

Figure 14: The base component with configurable icon, parameter value, unit, and trend indicator, as well as sample configurations for heart rate, tool help, and countdown.

The icons replace labels and should therefore be as unambiguous as possible. In some cases, the unit helps identify the value (e.g., degrees Fahrenheit or Celsius).

If this is not the case, they should preferably be omitted.

Trend indicators are displayed based on the difference between the current and previous values. An unusually sharp increase or decrease in the value may indicate events that warrant attention. For dynamic elements such as these, the frequency of changes should not be too high to avoid distractions.

All of these values are displayed continuously, with the exception of the time display, which appears only after a certain period of time has elapsed. The appearance of this element in the peripheral vision signals (for example) that half of the time limit has passed, which encodes this action itself with information.

Since the working time is limited by various parameters and is therefore not fixed, the component is formatted as a countdown. To avoid drawing attention to itself with too frequent changes, the display shows only minutes rather than counting down to the exact second.

8.1.3 Notifications and Pop-Ups

Referencing conventional industrial warning signs, the hazard icons were created in a triangular and a diamond-shaped version. Comparing them however, reveals that the diamond shape has smaller overall dimensions, which also means there is less space available for the illustration.

To standardize their function, the same silhouette is used for all icons, meaning that their spatial arrangement is much more important for peripheral distinction. Similar to the warning lights on a car's dashboard, each symbol has a fixed position to facilitate rapid information acquisition.

The prime example of the notification system is the high-priority, billboard-style alert (see Figure 8). The large, bright area ensures good visibility even when the pop-up is not in the center of the user's field of view. Since it has the highest priority, it is the only element placed in the upper central area of the screen. To avoid obscuring vision for too long, the pop-up shrinks to a smaller version but continues to flash to remain in the user's awareness.

This occurs without user intervention. Low oxygen levels or hazard proximity do require action, but in the intended scenarios there may be no opportunity to act immediately, and the HUD should not hinder the user's decision-making, which would be the case with a persistent alarm.

Figure 15: Hazard indicator icons for electricity, explosive atmosphere and fire. The triangle and diamond shapes reference warning and hazard signage, respectively.

8.1.4 The Checklist Component

Since the checklist is a text-based element, it is not intended to be comprehensible without looking at it. Nevertheless, some of the previous guidelines can be applied. To draw the user's attention to the current task, it is vertically centered and stands out more clearly from the other tasks on the screen.

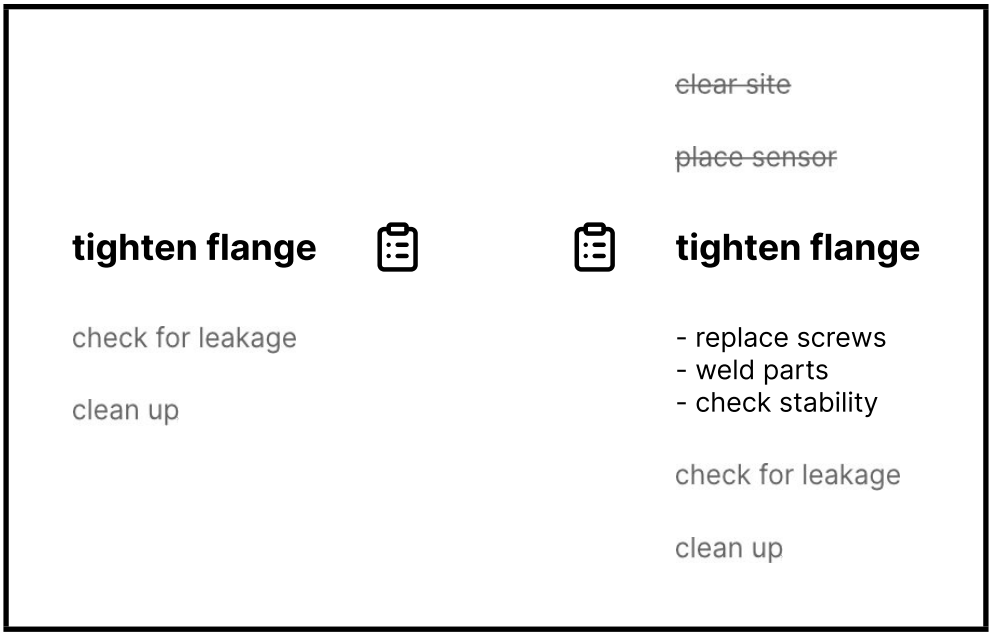

Figure 16: Default and expanded checklist. Note the center alignment of the current task to avoid losing focus when the list expands.

The icon serves as an anchor for potential gaze-based interaction, for example, to

display additional details such as completed tasks and subtasks. The design keeps the current task centered even in the detail view to enable faster orientation. This also mitigates the impact of potential false alarms—i.e., if the system mistakenly opens the detail view.

8.2 UI Examples

Note: the following illustrations are not to scale.

The full-featured version of the user interface is based on Microsoft's HoloLens 2, which has a horizontal field of view of 43°, a resolution of 1440 × 936 pixels per eye, and covers the entire color spectrum and provides full color availability (within the technical limits mentioned in Section 3).

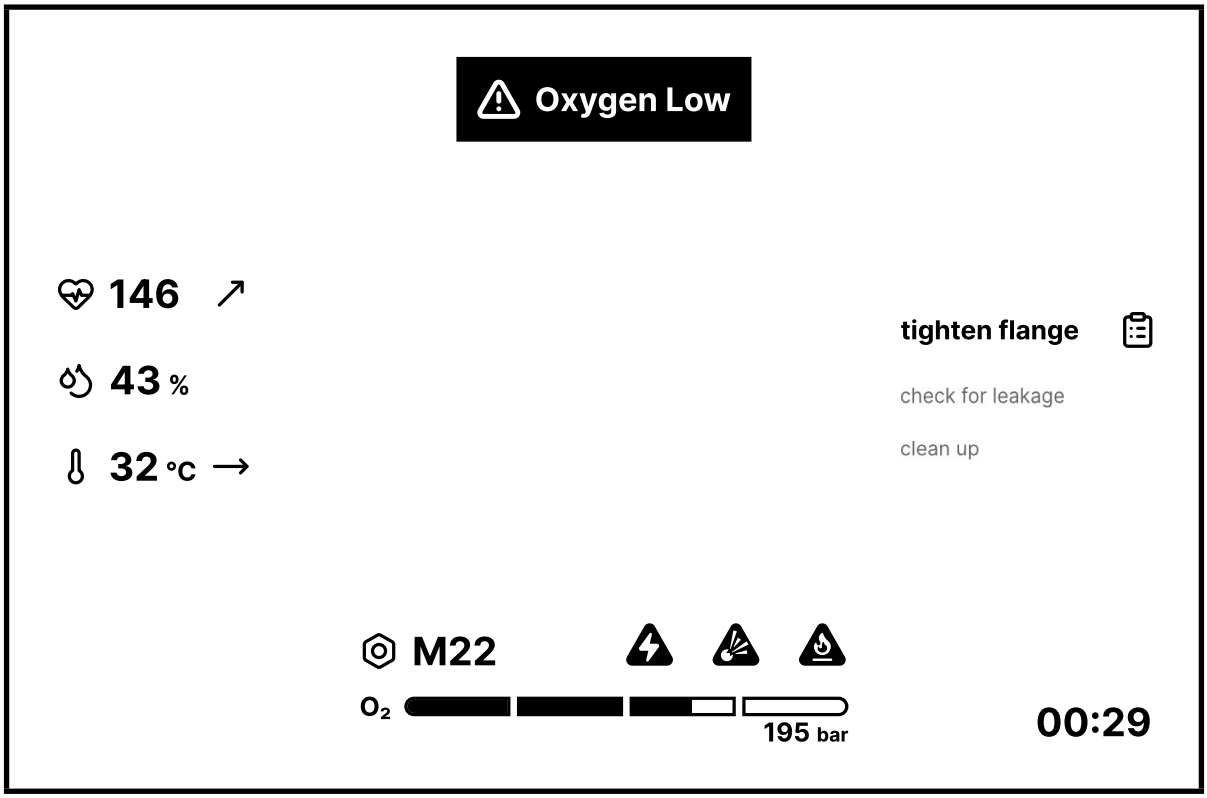

Figure 17: Full-featured UI concept for a binocular HUD.

Although the oxygen component is the most important, it is not necessarily the one that is checked most often—however, it should be the easiest to check. Therefore, it is placed in the lower center. Occasional information pop-ups are placed above the oxygen bar for the same reason: looking down is faster than looking up.

The block containing individual parameters is located on the left side, so that the status indicators are seen first, since they are the primary information of this component. This leaves the right side free for the checklist, which provides visual guidance thanks to its left-aligned text.

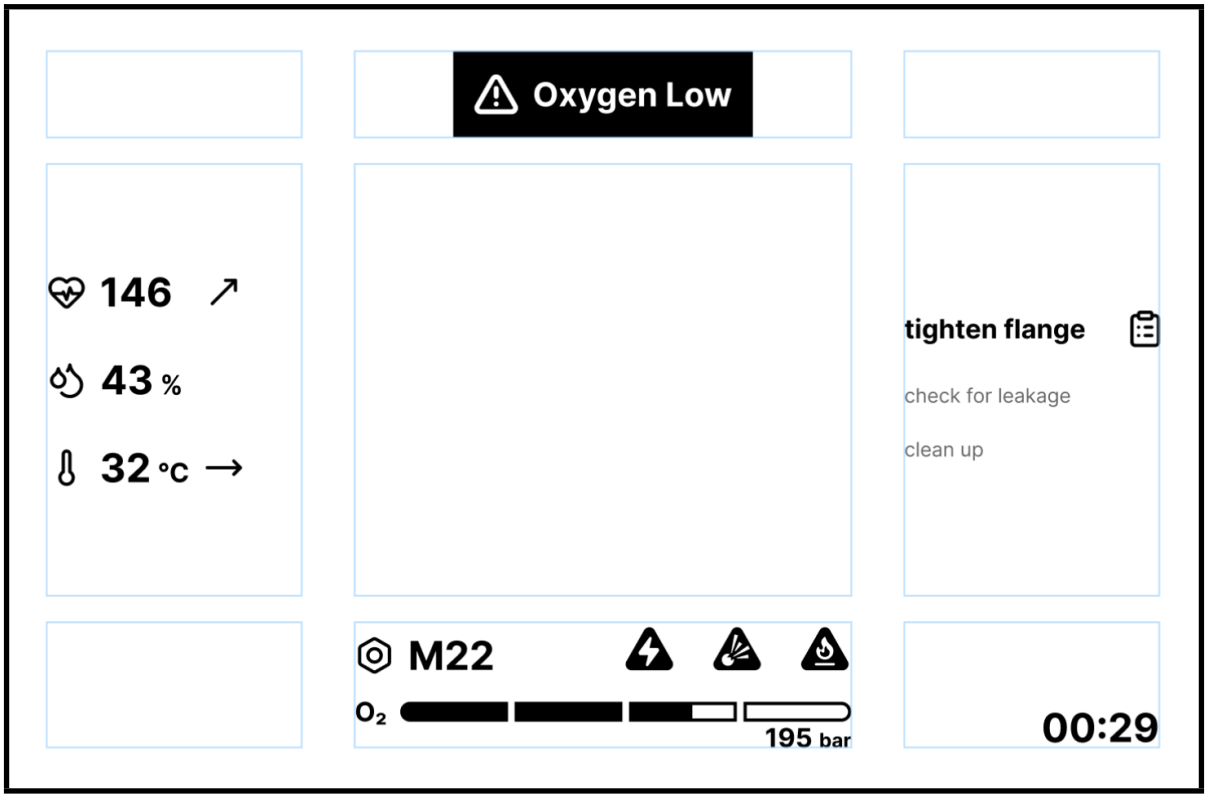

Figure 18: Grid guide of the full-featured UI

The minimal version of the user interface is based on the Google Glass Enterprise Edition 2, which features a monocular display with a resolution of 640 × 360 pixels. The limited space requires strict prioritization of elements. To ensure that the information is still easy to take in, all elements should be carefully aligned.

Arguably, some overlap of components would be useful to make better use of the limited space. However, this would disrupt the grid and visual hierarchy, decreasing visual search performance.

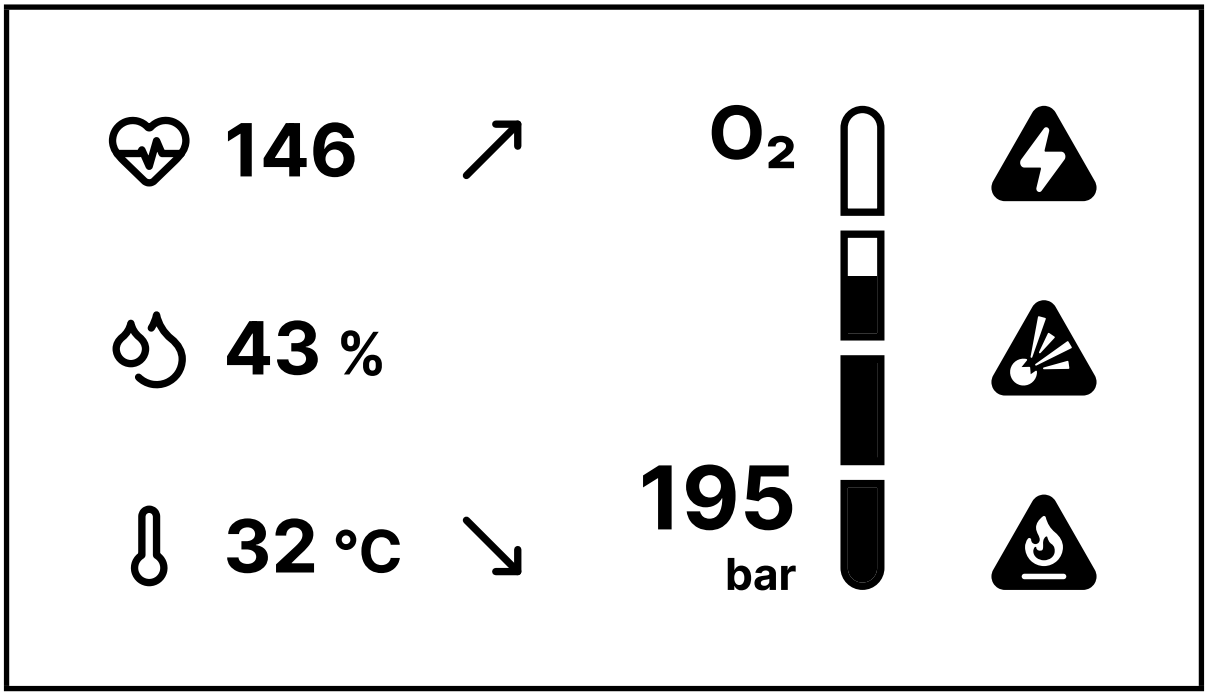

Figure 19: UI concept for a monocular HUD, worn in the upper left of the field of vision.

Figure 20: A negative example of overlapping elements. Placing the hazard symbols below the oxygen bar would save space, but disrupts the visual hierarchy.

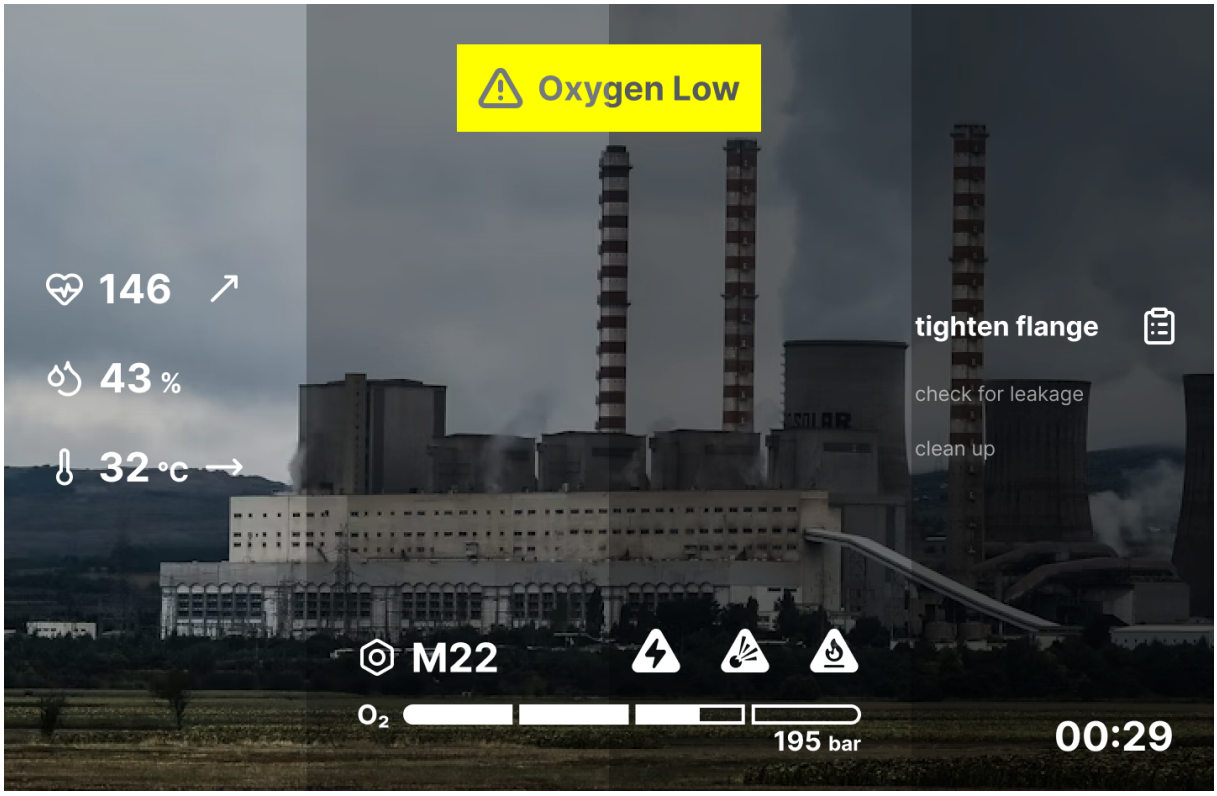

Figure 21: Full-featured HUD. Four black layers with 25% opacity have been added and stacked on top of each other to visualize lenses with varying degrees of tint often added by the manufacturers.

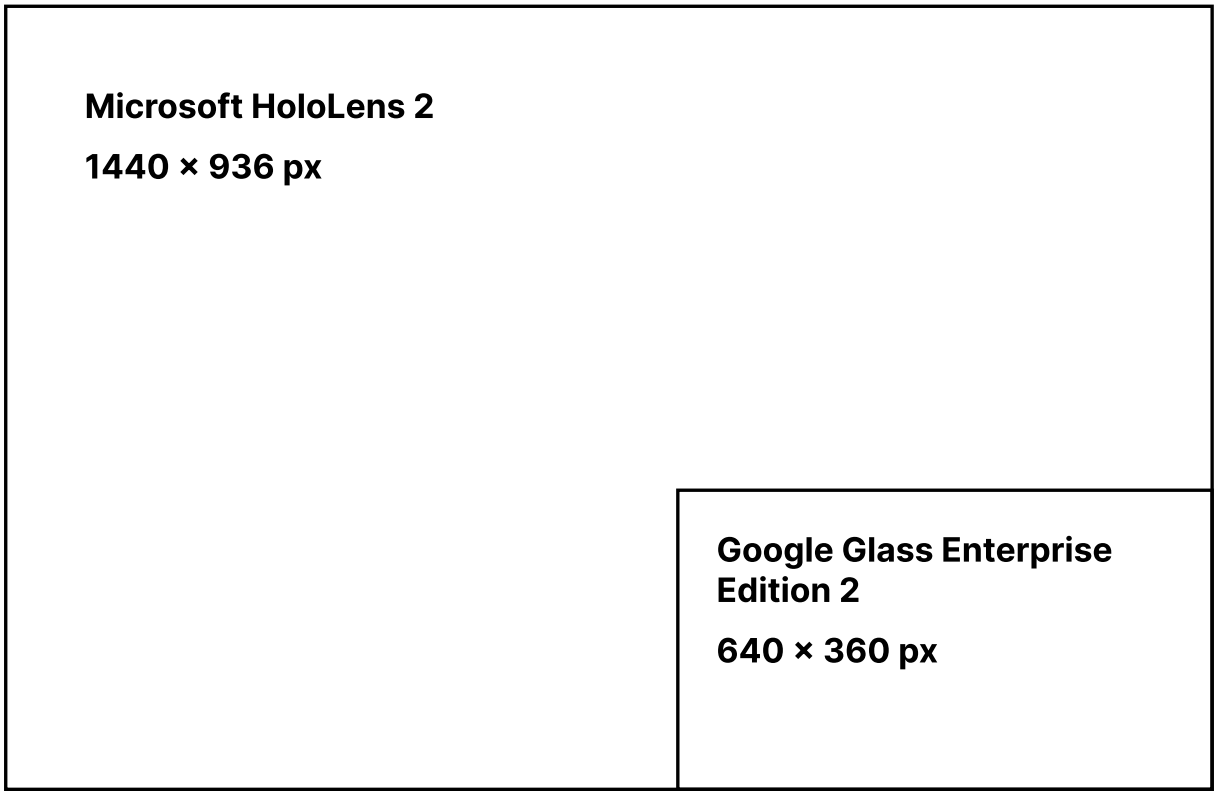

Figure 22: Size comparison of the Microsoft HoloLens 2 and the Google Glass Enterprise Edition 2

9 Conclusion

This project demonstrates the successful development of UI concepts for potential AR technologies intended for use in personal protective equipment, including a detailed presentation of sample components.

The provided design guidelines offer a collection of relevant concepts intended to serve as a solid foundation for exploring further industrial or safety-critical applications that could benefit from head-mounted HUDs. This reinforces the classification of 2D HUDs as augmented reality, as they serve as tools for applications that cannot be covered by world-stabilized systems.

The next step in further developing this topic is the creation of a prototype to evaluate the presented concepts. Technological limitations of OHMDs could be circumvented by testing aspects of the interfaces using video see-through systems. This would make it possible to test interaction concepts—such as the head-glance interface—as well as animations for different notification levels.

Furthermore, multimodality could be explored, as audio in particular could significantly enhance many of the concepts. Progress toward a completely hands-free system should also not be overlooked.

Appendix

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Dresden, den 14.04.2026

Lloyd König